Page a day worksheet **PAD nr: 4**

Grade 10 – Graphs and Equations – 1

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $ax + b = cx + 2$ (3)

(b) $5^{x-1} = \left(\frac{1}{25}\right)^{x+2}$ (3)

[6]

QUESTION 2

Given: $g(x) = \left(\frac{1}{3}\right)^x - 9$

(a) Calculate the intercepts with the axes. (4)

(b) Make a neat sketch of the graph below. (4)

(c) Give the range of the graph. (2)

(d) Give the equation of the graph if $g(x)$ is translated 3 units up. (2)

[12]

Page a day worksheet **PAD nr: 5**

Grade 10 – Graphs and Equations – 2

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

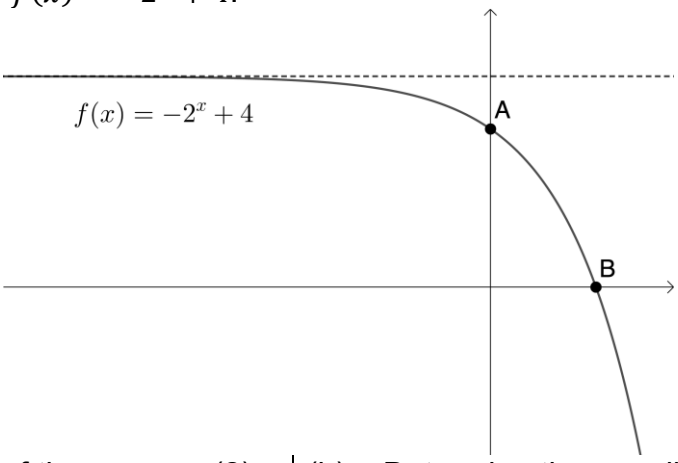
(a) $(x-2)(2x+5)=0$ (2)

(b) $0{,}125-\left(\frac{1}{2}\right)^{x+2}=0$ (4)

[5]

QUESTION 2

The graph below shows $f(x)=-2^x+4$.



(a) Give the equation of the asymptote. (2)

(b) Determine the coordinates of A and B. (3)

(c) Give the equation reflection of the graph in the y-axis. (2)

(d) Give the range of the graph. (2)

[9]

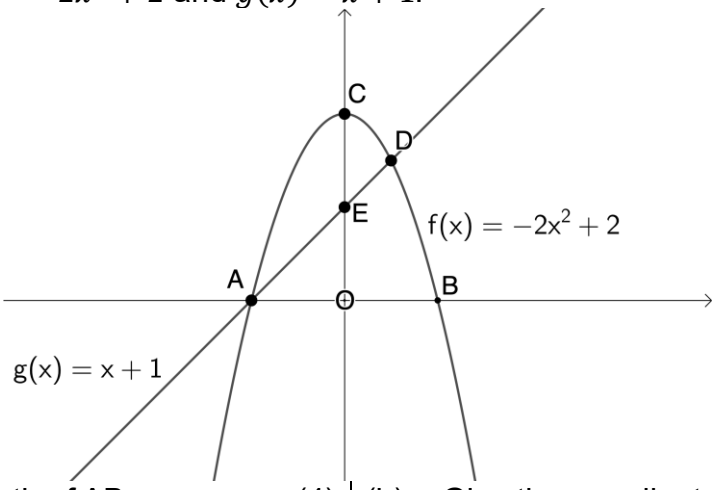
Page a day worksheet **PAD nr: 6**

Grade 10 – Graphs and Equations – 3

Name:	

QUESTION 1

The graph shows $f(x) = -2x^2 + 2$ and $g(x) = x + 1$.



(a) Determine the length of AB. (4)

(b) Give the coordinates of E. (2)

(c) Give the range of $f(x)$ (2)

(d) Give the equation of $f(x)$ reflected in the x-axis. (2)

(e) For which values of x will $f(x)$ be increasing? (2)

(f) Describe the transformation of $g(x)$ to $y = -x + 1$ (2)

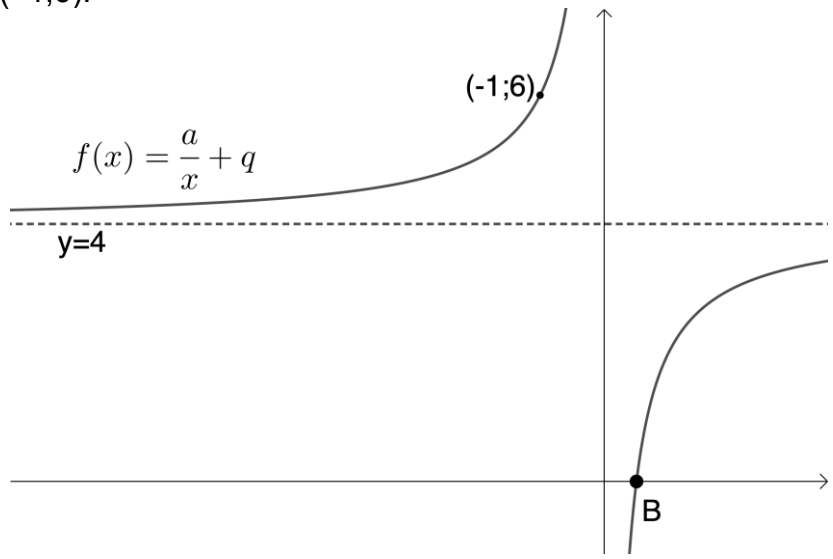
[14]

Page a day worksheet **PAD nr: 7**
Grade 10 – Graphs and Equations – 4

Name:	

QUESTION 1

The graph shows $f(x) = \frac{a}{x} + q$ which has a horizontal asymptote at $y = 4$ and passes through the coordinates of (–1;6).



(a) Prove that the values of $a = -2$ and $q = 4$. (4)

(b) Determine the coordinates of B. (3)

(c) Give the domain and range of $f(x)$. (2)

(d) Give the equation of the axis of symmetry with a positive gradient. (2)

(e) Determine the equation of the reflection of $f(x)$ in the y-axis. (2)

[14]